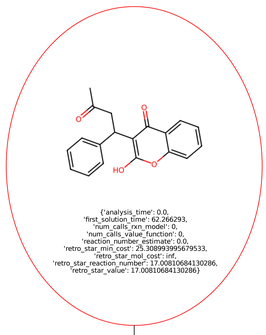
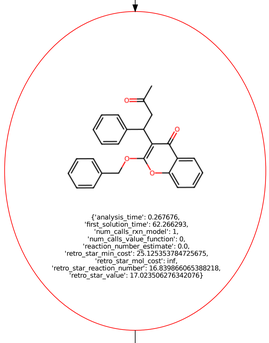


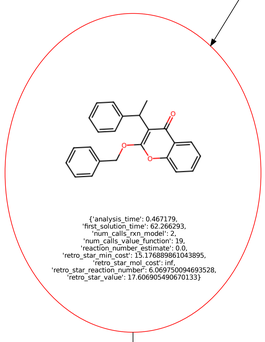
{'analysis_time': 0.0,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 25.308993995679533,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 17.00810684130286,
'retro_star_value': 17.00810684130286}

Reaction
{'analysis_time': 0.267676,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 25.308993995679533,
'retro_star_reaction_number': 17.023506276342076,
'retro_star_rxn_cost': 0.18364021095385843,
'retro_star_value': 17.023506276342076}

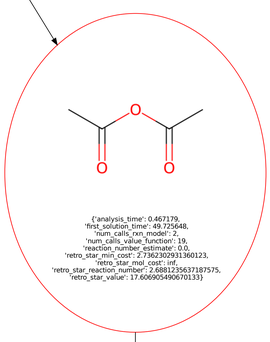


{'analysis_time': 0.267676,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 25.125353784725675,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 16.839866066388218,
'retro_star_value': 17.023506276342076}

Reaction
{'analysis_time': 0.467179,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 2,
'num_calls_value_function': 19,
'retro_star_min_cost': 26.578511775483896,
'retro_star_reaction_number': 17.423565279716276,
'retro_star_rxn_cost': 8.66539162130399,
'retro_star_value': 17.606905490670133}



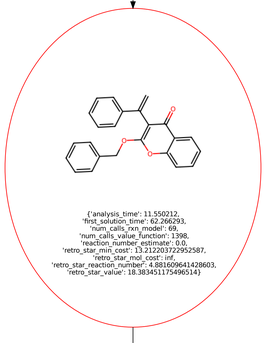
{'analysis_time': 0.467179,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 2,
'num_calls_value_function': 19,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 15.176889861043895,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 6.069750094693528,
'retro_star_value': 17.606905490670133}



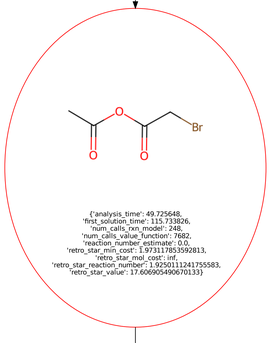
{'analysis_time': 0.467179,
'first_solution_time': 49.725648,
'num_calls_rxn_model': 2,
'num_calls_value_function': 19,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 2.7362302931360123,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 2.6881235637187575,
'retro_star_value': 17.606905490670133}

Reaction
{'analysis_time': 11.550212,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 69,
'num_calls_value_function': 1398,
'retro_star_min_cost': 15.176889861043895,
'retro_star_reaction_number': 6.846295779519 91,
'retro_star_rxn_cost': 1.9646861380913074,
'retro_star_value': 18.383451175496141}

Reaction
{'analysis_time': 49.725648,
'first_solution_time': 115.733826,
'num_calls_rxn_model': 248,
'num_calls_value_function': 7682,
'retro_star_min_cost': 2.7362302931360123,
'retro_star_reaction_number': 2.6881235637187575,
'retro_star_rxn_cost': 0.7631124395431991,
'retro_star_value': 17.606905490670133}



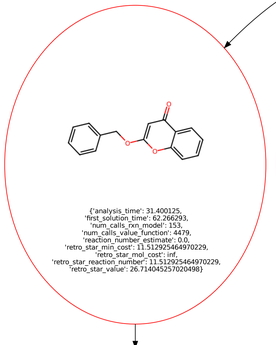
{'analysis_time': 11.550212,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 69,
'num_calls_value_function': 1398,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 13.212203722952587,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 4.881609641428603,
'retro_star_value': 18.383451175496141}



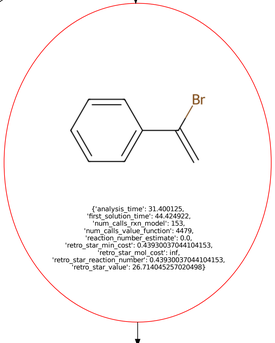
{'analysis_time': 49.725648,
'first_solution_time': 115.733826,
'num_calls_rxn_model': 248,
'num_calls_value_function': 7682,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 1.9731178535592813,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 1.9250111241755583,
'retro_star_value': 17.606905490670133}

Reaction
{'analysis_time': 31.400125,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 153,
'num_calls_value_function': 4479,
'retro_star_min_cost': 13.212203722952587,
'retro_star_reaction_number': 13.212203722952587,
'retro_star_rxn_cost': 1.2599778875413183,
'retro_star_value': 26.714045257020498}

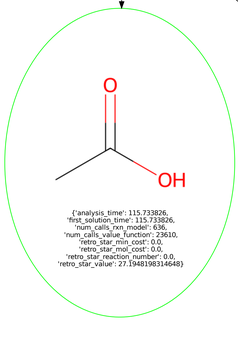
Reaction
{'analysis_time': 115.733826,
'first_solution_time': 115.733826,
'num_calls_rxn_model': 636,
'num_calls_value_function': 23610,
'retro_star_min_cost': 11.512925464970229,
'retro_star_reaction_number': 11.512925464970229,
'retro_star_rxn_cost': 11.512925464970229,
'retro_star_value': 27.194819831464805}



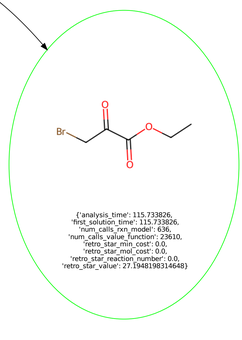
{'analysis_time': 31.400125,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 153,
'num_calls_value_function': 4479,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 11.512925464970229,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 11.512925464970229,
'retro_star_value': 26.714045257020498}



{'analysis_time': 31.400125,
'first_solution_time': 44.424922,
'num_calls_rxn_model': 153,
'num_calls_value_function': 4479,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.4393003704410415 3,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 0.4393003704410415 3,
'retro_star_value': 26.714045257020498}



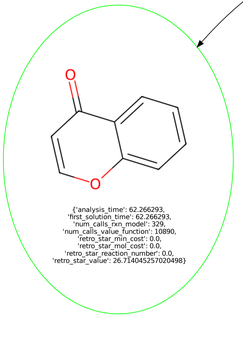
{'analysis_time': 115.733826,
'first_solution_time': 115.733826,
'num_calls_rxn_model': 636,
'num_calls_value_function': 23610,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 27.194819831464805}



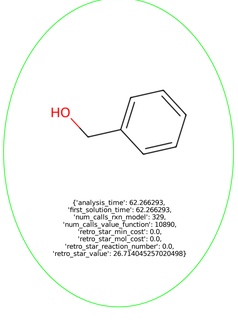
{'analysis_time': 115.733826,
'first_solution_time': 115.733826,
'num_calls_rxn_model': 636,
'num_calls_value_function': 23610,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 27.194819831464805}

Reaction
{'analysis_time': 62.266293,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 329,
'num_calls_value_function': 10890,
'retro_star_min_cost': 11.512925464970229,
'retro_star_reaction_number': 11.512925464970229,
'retro_star_rxn_cost': 11.512925464970229,
'retro_star_value': 26.714045257020498}

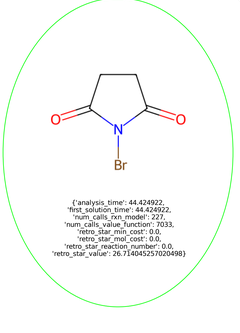
Reaction
{'analysis_time': 44.424922,
'first_solution_time': 44.424922,
'num_calls_rxn_model': 227,
'num_calls_value_function': 7033,
'retro_star_min_cost': 0.4393003704410415 3,
'retro_star_reaction_number': 0.4393003704410415 3,
'retro_star_rxn_cost': 0.4393003704410415 3,
'retro_star_value': 26.714045257020498}



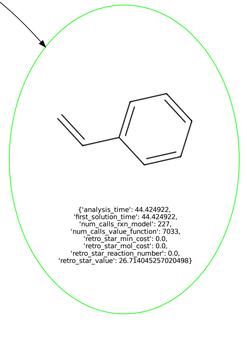
{'analysis_time': 62.266293,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 329,
'num_calls_value_function': 10890,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 26.714045257020498}



{'analysis_time': 62.266293,
'first_solution_time': 62.266293,
'num_calls_rxn_model': 329,
'num_calls_value_function': 10890,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 26.714045257020498}



{'analysis_time': 44.424922,
'first_solution_time': 44.424922,
'num_calls_rxn_model': 227,
'num_calls_value_function': 7033,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 26.714045257020498}



{'analysis_time': 44.424922,
'first_solution_time': 44.424922,
'num_calls_rxn_model': 227,
'num_calls_value_function': 7033,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 26.714045257020498}